



Container Security

From Basics to Advanced Protection

A comprehensive, hands-on course for securing containerized environments



3.5-4 Hours |  12+ Hands-On Labs

Course Overview

Container Security: From Basics to Advanced Protection is a practical, hands-on course designed to equip developers, DevOps engineers, and security professionals with the knowledge and skills to secure containerized applications. This course goes beyond theory, providing real-world attack demonstrations, defensive techniques, and production-ready security configurations.

6

Modules

20+

Video Lessons

12

Practical Labs

Who Should Take This Course?

- **DevOps Engineers** managing containerized infrastructure
- **Security Professionals** expanding into cloud-native security
- **Software Developers** building containerized applications
- **System Administrators** deploying container platforms
- **Platform Engineers** responsible for Kubernetes clusters

Prerequisites

- Basic Docker/container experience (running containers, basic commands)
- Linux command line familiarity

- Understanding of basic networking concepts
- Access to a Linux environment (VM or WSL2)

What You'll Learn

Module 1: Container Security Fundamentals

Duration: 30-35 minutes

Understand the core concepts of container isolation, the shared kernel reality, and fundamental attack surfaces.

✓ Namespaces & cgroups

✓ Privileged containers

✓ Container capabilities

✓ Exposed Docker API

Key Takeaway: Containers are NOT virtual machines—they share the host kernel, making isolation mechanisms critical for security.

Module 2: Container Attack Vectors and Vulnerabilities

Duration: 35-40 minutes

Explore real-world attack scenarios including container escapes, image tampering, and resource exhaustion attacks.

✓ Container escape techniques

✓ Kernel exploits

✓ Malicious images

✓ DoS attacks

✓ Content Trust

✓ Insecure configurations

Key Takeaway: Attackers can exploit misconfigurations, vulnerable images, and kernel weaknesses to break out of containers—defense requires multiple layers.

Module 3: Seccomp Profiles in Practice ★

Duration: 40-45 minutes

Master system call filtering with Seccomp—from understanding defaults to creating custom profiles for production applications.

✓ Seccomp fundamentals

✓ Default Docker profiles

✓ Custom profile creation

✓ Using strace

✓ Debugging denials

✓ Production best practices

Key Takeaway: Seccomp provides kernel-level syscall filtering that dramatically reduces attack surface—custom profiles tailored to applications offer the strongest protection.

Module 4: Mandatory Access Control - AppArmor and SELinux

Duration: 30-35 minutes

Implement Mandatory Access Control systems to enforce fine-grained security policies beyond traditional permissions.

✓ MAC vs DAC models

✓ AppArmor profiles

✓ SELinux contexts

✓ Policy generation

✓ Troubleshooting denials

✓ Tool selection

Key Takeaway: MAC systems like AppArmor and SELinux enforce policies that restrict processes even when running as root—essential for defense in depth.

Module 5: Orchestration and Supply Chain Security

Duration: 30-35 minutes

Secure Kubernetes deployments and establish supply chain controls including image scanning, signing, and runtime monitoring.

✓ Pod Security Standards

✓ Security contexts

✓ Network policies

✓ Image scanning (Trivy/Grype)

✓ Image signing (Cosign)

✓ Runtime monitoring (Falco)

Key Takeaway: Kubernetes security requires policy enforcement, supply chain validation, and continuous runtime monitoring to prevent and detect threats.



Hands-On Lab Highlights

- **Privileged Container Escape** - Break out of a container and access the host
- **Image Forensics** - Analyze a compromised container image layer-by-layer
- **Custom Seccomp Profiles** - Build syscall filters for nginx and Python apps
- **AppArmor Profile Generation** - Create and enforce custom security policies
- **Pod Security Standards** - Lock down Kubernetes with restrictive policies
- **Automated Image Scanning** - Integrate Trivy into CI/CD pipelines

- **Runtime Threat Detection** - Deploy Falco and analyze security events

Course Approach & Methodology



100% Practical, Zero Fluff

Every concept is immediately followed by a demonstration or hands-on lab. No PowerPoint-only lectures—you'll see attacks in action and implement defenses yourself.

Content Mix:

- **35-40% Terminal/CLI:** Live command execution, exploit demonstrations, log analysis
- **20-25% Code Editor:** Writing Seccomp profiles, YAML configurations, policy files
- **20-25% Slides:** Architecture diagrams, threat models, security concepts
- **15-20% Browser:** Documentation, scanning dashboards, tool interfaces

Progressive Complexity: Start with fundamentals, build to advanced techniques. Each module builds on previous knowledge, creating a cohesive learning path.

Real-World Focus: All examples based on actual CVEs, production scenarios, and documented attack patterns. Learn what actually matters in production environments.

Technical Requirements



Lab Environment

- **RAM:** 4GB minimum, 8GB recommended
- **Disk:** 20GB free space
- **OS:** Linux (Ubuntu 20.04+ or similar) or WSL2 on Windows
- **Software:** Docker Engine 24.0+, kubectl, strace, auditd
- **Optional:** Kubernetes cluster (minikube/kind acceptable)

All necessary vulnerable containers, sample profiles, and validation scripts are provided.

Learning Outcomes

By the end of this course, you will be able to:

-  **Identify** container security vulnerabilities in existing deployments
-  **Exploit** common container weaknesses (in lab environments)
-  **Create** custom Seccomp, AppArmor, and SELinux profiles
-  **Implement** Pod Security Standards in Kubernetes
-  **Integrate** image scanning and signing into CI/CD pipelines
-  **Deploy** runtime monitoring and threat detection systems
-  **Design** layered security architectures for containerized applications
-  **Troubleshoot** security policy denials and misconfigurations
-  **Assess** risk and prioritize security controls for your environment

Beyond the Course

Container security is an evolving field. This course provides not just current best practices, but the foundational knowledge to adapt as new threats emerge. You'll learn:

- How to stay current with container security research
- Resources for ongoing learning (blogs, tools, communities)
- Frameworks for evaluating new security tools
- Strategies for implementing security incrementally in existing environments

Assessment & Certification

The course includes:

- **18 module-level quizzes** (3 MCQs per module)
- **12 hands-on labs** with automated validation
- **1 comprehensive final assessment** (10 MCQs + capstone lab)
- **Passing criteria:** 75% on assessments + all labs completed

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